

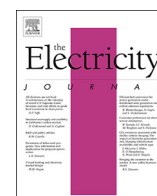


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Modelling and management of smart microgrid for rural electrification in sub-saharan Africa: The case of Nigeria



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ARTICLE INFO

Keywords:

Energy access
Off-grid rural electrification
HOMER
Simulink
Nigeria

ABSTRACT

Access to electricity is still a challenge in many parts of sub-Saharan Africa. In Nigeria, over 70% of the rural dwellers do not have access to electricity. The purpose of this paper is to examine the potential of a smart microgrid for off-grid rural electrification in Nigeria. A combination of design thinking and model-based design methodology is employed to select a suitable microgrid configuration and to develop a smart microgrid model. A system consisting of a solar photovoltaic array, battery energy storage and a diesel generator is selected, and the model is developed in Simulink. Demand data from 10 rural communities in Nigeria are used to validate the performance of the model and the potential for demand management is considered. The use of energy efficient light bulbs is found to reduce the peak electricity demand of the case study communities by 42 to 76%. Combining the proposed system with the use of LED bulbs makes the system to have 56 to 81% less net present cost than a system with a diesel generator alone and incandescent light bulbs. The proposed smart microgrid is found to be more suitable for off-grid rural electrification in Nigeria than diesel generators which are currently used for off-grid electrification in Nigeria.

1. Introduction

Access to electricity continues to elude many households and around 588 million persons in SSA do not have access to electricity (IEA, 2017). In 2011, the United Nations launched a “Sustainable Energy for All” initiative. At the heart of this initiative is a call for universal access to affordable, reliable and modern energy services by 2030. The initiative also includes a target to double the rate of improvement in energy efficiency and increase substantially the share of renewable energy in the global energy mix. It must be noted that the vast majority of the people in SSA who lack access to electricity live in rural areas (World Bank, 2017). Therefore, in order to achieve universal access to modern energy services, rural electrification must be made a

priority.

Nigeria is the most populous and largest economy in Africa, endowed with a plethora of energy resources but about 70% of its rural population do not have access to electricity (IEA, 2017). The country is currently struggling to provide electricity for all its citizens. The Nigerian electricity sector is faced with several challenges. The transmission network is currently overloaded due to poor maintenance and vandalization. It experiences losses of up to 25% (Energylopedia, 2017). The radial network is unreliable and experiences frequent system collapse. The distribution network also suffers from technical losses, vandalization, electricity theft, poor maintenance and lack of functional automatic control system. Due to regular black-outs, consumers are forced to purchase stabilizers for the protection of electronic

Abbreviations: AC, alternating current; AMI, advanced metering infrastructure; CAES, compressed air energy storage; CFL, compact fluorescent lamp; DC, direct current; DISCO, distribution company; DSN, distributed Sensor Network; EU, European Union; GHI, global horizontal irradiance; HOMER, hybrid optimisation model for electric renewables; ICT, information and communication technology; IDS, intrusion detection systems; IEA, international energy agency; kW, kilowatts; L, litre; LCOE, levelized cost of electricity; LEC, levelized energy cost; LED, light emitting diodes; LOLP, loss of load probability; Mtoe, metric tonnes of oil equivalent; MW, megawatts; NPC, net present cost; PHCN, power holding company of Nigeria; PV, photovoltaic; RC-SHP, regional centre for small hydro power in Africa; REA, rural electrification agency; SOC, State of charge of battery; SSA, sub-saharan Africa; STC, standard test condition; UNIDO, united nations industrial development organisation; WSN, wireless sensor network

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<https://doi.org/10.1016/j.tej.2019.106672>

Available online 11 October 2019

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appliances. Only about 59% of Nigeria's population is connected to the electricity grid (World Bank, 2018). Majority of people who are not connected to the central grid live in rural areas. Thus, to provide electricity to those living in rural parts of Nigeria, microgrid solutions will play a key role.

In literature, there exist several studies relating to microgrid development. For examples, techno-economic feasibility of rural electrification projects has been assessed in (Reddy et al., 1990). (Borowy and Salameh, 1996) made use of the graphical construction technique to determine the optimal combination of a photovoltaic array and battery in a hybrid PV-wind system. (Kaabeche et al., 2011) presented sizing optimisation model for hybrid PV, wind and battery storage using an iterative technique with Loss of load probability and LEC as optimisation objectives. (Hocaglu et al., 2009) used a similar iterative optimization methodology but assumed that the battery capacity was infinite. (Nandi and Ghosh, 2009) used the HOMER software for optimal sizing of a hybrid PV, wind and battery system. There are some research works focused on developing new optimization methodologies for sizing of system components (Perera et al., 2013; Ou and Hong, 2014; Mellit, 2009; Mohamed and Eltamaly, 2018). From the literature, we observed that although there has been a lot of focus on microgrid systems for rural electrification in developing countries such as SSA countries, the application of smart grid features such as demand-side management has been very little. Therefore, there is a knowledge gap that needs to be filled in this area.

The aim of this paper is to investigate the potential of a smart microgrid for rural electrification in Nigeria. To this end, the paper assesses the renewable energy resources in Nigeria and decides a system configuration suitable for most rural areas. Next, a smart microgrid model is developed and the electricity demand for case study rural communities in Nigeria is determined. Finally, the smart microgrid components are optimised and simulations and system performance studies are carried out. Although there are a number of hybrid-microgrid researches for the study country Nigeria, and sub-Saharan Africa in general, there has been no record of previous smart microgrid research. Furthermore, due to the difficulty of obtaining real demand data, the previous works for Nigeria including (Olatomiwa et al., 2015) and (Albert et al., 2015) used assumed demand data. The current study utilized actual load data from Nigeria and hence was able to consider the potential for demand shifting and analyse the impact of smart grid features such as energy efficient appliances on system size, performance and reliability.

2. Approach and methodology

This study employed a combination of design thinking and model-based design methodology.

2.1. Design thinking

Design thinking focuses on developing solutions that are not only technically and economically feasible but also suitable for the context for which they are developed. It is an important aspect of this research because it ensures that the solutions proposed are indeed suitable for the Nigerian rural electrification context. Design thinking is categorised into three steps namely, understand, explore and materialise. Methods for design thinking includes interviews, examining existing solutions, creating prototypes, mind mapping, asking relevant questions, and situational analysis.

2.2. Model-based design

Model-based design is a mathematical and visual method of developing complex control systems. It is the use of models for system design, analysis, simulation and verification of results (Prinsloo and Dobson, 2015). The model-based design supports a variety of research activities

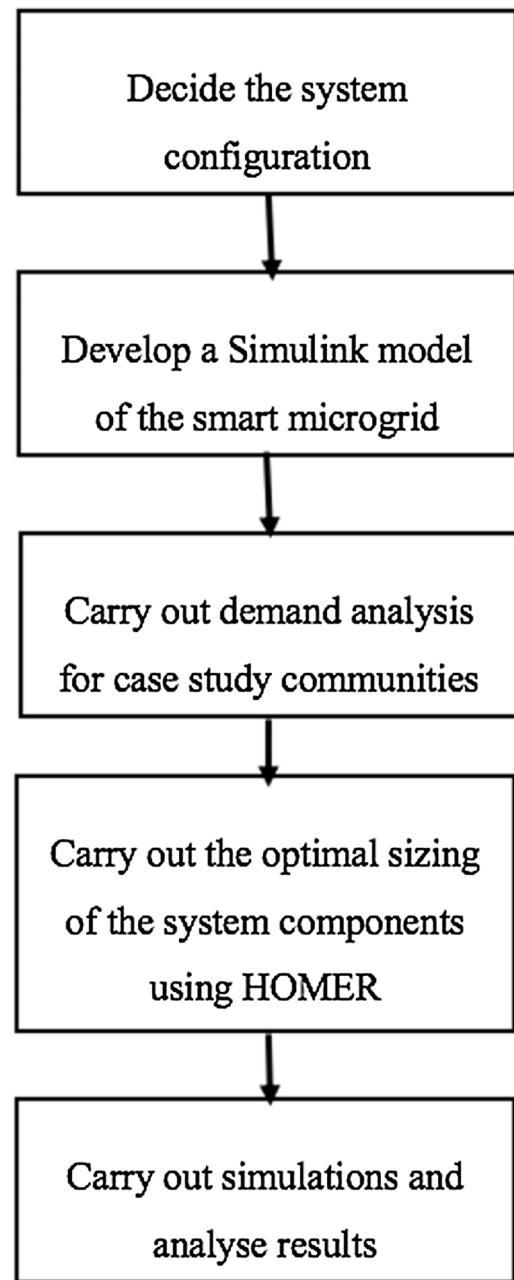


Fig. 1. General methodological framework.

such as technology selection, system integration, system testing, optimisation, cost-benefit analysis and sensitivity analysis. Its application is seen in solving engineering problems such as controls, signal processing and communication systems. This research combined field survey data obtained from the UNIDO in Nigeria and desk-based analysis. Fig. 1 shows the general methodological framework used in this study.

2.3. System configuration and modelling

2.3.1. Renewable energy resources in Nigeria

Nigeria average daily global horizontal solar irradiation ranges from 4.2 kWh/m² in the south to 6.2 kWh/m² in the north. The country experiences an average of 6 h of sunshine daily (Solargis, 2017). Nigeria has moderate wind energy potential. The country experiences an annual average wind speed of about 2 m/s in the coastal areas and 4 m/s in the far northern region at a hub height of 10 m (Africa-EU renewable energy cooperation, n.d.). The potential for small hydropower in

Nigeria is estimated to be 3500 MW but only 64.2 MW has been exploited (Africa-EU renewable energy cooperation, n.d.). The potential for large hydropower is estimated to be 11,250 MW with an exploited capacity of only 1900 MW. Nigeria's biomass resources include crops, forage grasses, shrubs and wastes. The daily production of animal waste is estimated to be 227,500 tonnes.

2.3.2. System configuration

From the review of renewable energy resources highlighted above, it is evident that Nigeria has significant solar potential in all parts of the country. Solar based microgrid system is therefore suitable. A microgrid system configuration consisting of solar photovoltaic (PV) array, battery storage and diesel generator was chosen. A backup energy source is required in order to ensure a reliable supply of electricity even during extended periods of low power generation. A diesel generator is convenient because it is the common means of off-grid generation currently used in Nigeria. Apart from the main load which represents the electricity demand of the community, a dummy load is included to dispose of excess power generated from the solar PV when the battery is fully charged. An inverter was included in the system to convert direct current from the PV and battery to alternating current used by the loads. Fig. 2 shows the system configuration.

2.3.3. Mathematical modelling of system components

High-level models similar to those used in (Mohamed and Eltamaly, 2018) and (Olatomiwa et al., 2015) were used in this study. This means that low-level interactions of current and voltages were not considered. The models for maximum power point tracking and operation of the inverter are not within the scope of this study. The output power of the solar PV, the state of charge of the battery and the fuel consumption of the diesel generator are considered.

Model for Solar PV module is given by Eq. (1).

$$P_{pv-out} = P_{pv-rated} \times \frac{G}{G_{ref}} \times [1 + K_T (T_c - T_{ref})] \quad (1)$$

Where P_{pv-out} is the power output of the solar PV, $P_{pv-rated}$ is the rated power of the PV system, G is the solar irradiance on a surface in W/m^2 , G_{ref} is the solar irradiance on a surface at standard test condition

($1000 W/m^2$), T_{ref} is the cell temperature at the reference condition ($25^\circ C$), K_T is the temperature coefficient of the PV module and for this study it was assumed to be -0.004 per $^\circ C$. T_c is the cell temperature and it is obtained from Eq. (2).

$$T_c = T_{amb} + (0.0256 \times G) \quad (2)$$

Where, T_{amb} is the ambient temperature and 0.0256 is the ross coefficient.

Model for battery storage is given by Eqs. (3) and (4).

Charging mode:

$$E_{B(t+1)} = E_{B(t)} (1 - \sigma) + \text{surplus energy from PV} \times \eta_{BC} \quad (3)$$

Discharging mode:

$$E_{B(t+1)} = E_{B(t)} (1 - \sigma) - \text{deficit energy} \times \eta_{BD} \quad (4)$$

Where $E_{B(t)}$ is the energy in the battery bank at time t , η_{BC} is the charging efficiency of the battery, η_{BD} is the discharging efficiency of the battery and σ is the battery self-discharge rate. For this study, the self-discharge rate was assumed to be zero. The charge and discharge efficiency of the battery were assumed to be both 80%.

The power supplied to the battery for charging is given by Eq. (5).

$$P_{BC(t)} = (P_{pv(t)} - P_{L(t)}) \times \eta_{BC} \quad (5)$$

The power supplied by the battery to the load is given by Eq. (6).

$$P_{BD(t)} = (P_{L(t)} - P_{pv(t)}) \times \eta_{BD} \quad (6)$$

Where $P_{BC(t)}$ is the power sent to the battery at time t , $P_{L(t)}$ is the power requirement of the load, $P_{pv(t)}$ is the power generated by the PV at time t and $P_{BD(t)}$ is the power delivered by the battery at time t .

At every time (t), the operation of the battery is guided by the constraint:

$$E_{Bmin} \leq E_{B(t)} \leq E_{Bmax} \quad (7)$$

Where E_{Bmin} is the minimum allowable storage capacity of the battery bank and E_{Bmax} is the maximum allowable storage capacity of the battery bank.

Diesel generator model is given by Eq. (8).

$$D_f(t) = \alpha_D P_{DG(t)} + \beta_D P_{DG-rated} \quad (8)$$

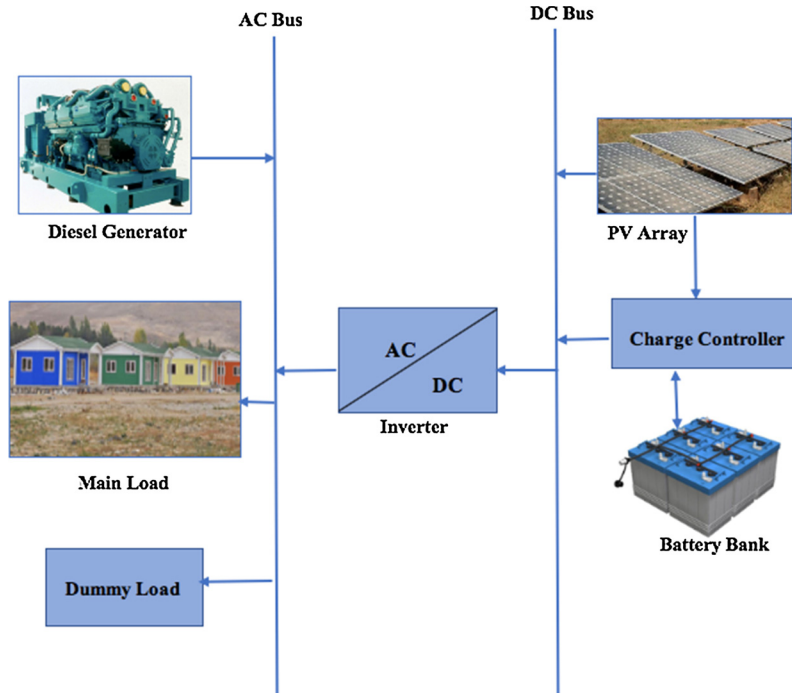


Fig. 2. Schematic diagram of solar PV, diesel generator and battery microgrid.

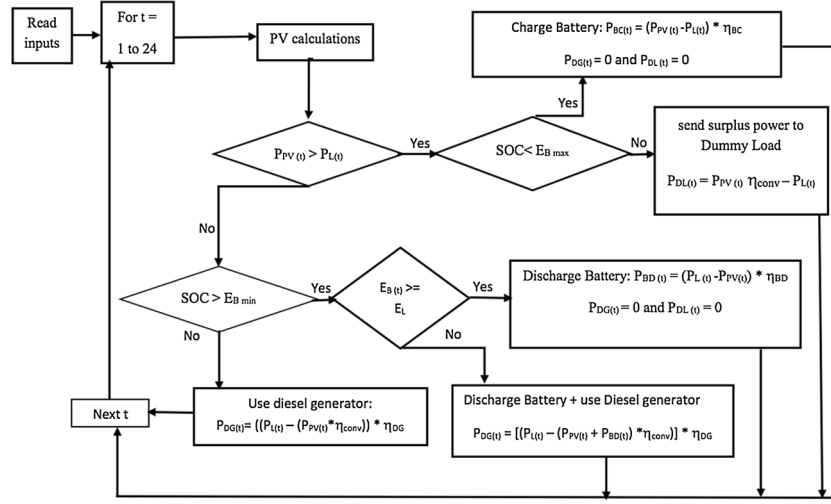


Fig. 3. Flowchart of the energy management model.

Where, $D_{f(t)}$ is the hourly fuel consumption of the diesel generator in L/h, $P_{DG(t)}$ is the output power of the diesel generator in kW, $P_{DG-rated}$ is the rated power of the diesel generator in kW, α_D and β_D are the coefficients of the fuel consumption curve in L/kWh. For this study, they were taken as 0.246 and 0.08145 respectively (Mohamed and Eltamaly, 2018).

2.3.4. Simulink model of energy management system

For this model, it is assumed that consumer appliances are smart and smart meters are used. The algorithm used in the model is shown in the flowchart in Fig. 3. All parameters are as previously defined. SOC is the battery state of charge, η_{conv} is the efficiency of the converter, η_{DG} is the efficiency of the diesel generator, E_L is the energy requirement of the load and $P_{DL(t)}$ is the power sent to the dummy load.

The operation of the converter (inverter) was not included in the Simulink model. However, it was assumed that the inverter was 95% efficient. Hourly electricity demand data, hourly solar irradiance data, hourly temperature, rated capacity of the PV array, rated capacity of the battery and the rated capacity of the diesel generator are the inputs for the model. The model uses these inputs to calculate the hourly power generated by the PV array, hourly state of charge of the battery, hourly power generated by the diesel generator and hourly power sent to the dummy load. Fig. 4 shows the outer layer of the Simulink model.

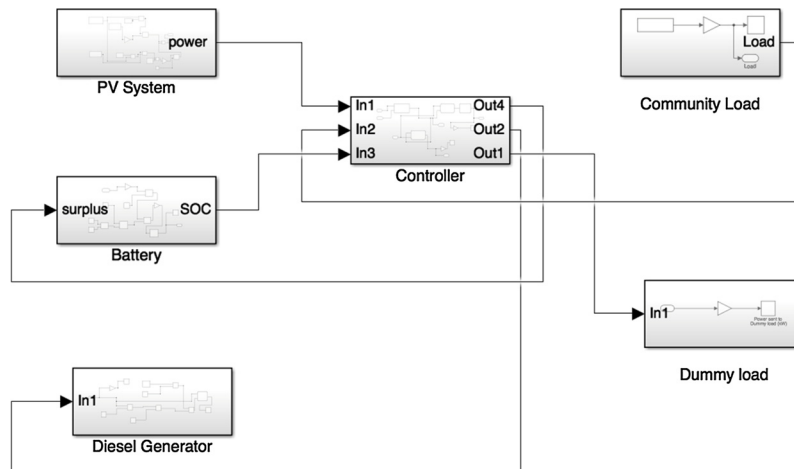


Fig. 4. Simulink model of the smart microgrid.

2.4. Optimal sizing of system components

2.4.1. Electricity demand for case study communities

The load data for the study communities were obtained from UNIDO in Nigeria. The RC-SHP located in Abuja Nigeria conducted a load survey for ten rural communities in Taraba state, namely: Sabon Gari, Kakara, Bangoba, Kusuku, Lugere, Galadima, Nguroje, Furmi, Tunga and Chabal Gudali. The load survey was carried out by UNIDO as part of the groundwork for the development of the organisation's Tunga Dam II small hydropower project in Nigeria. The load survey provided the peak electricity demand for activities in each of the rural communities. Table 1 shows the unit capacity for each of the activities and the total demands of the communities. The product of the unit capacity and the number of each activity gives the total demand.

2.4.2. Electricity demand profile

Hourly electricity demand is required for the analysis in this study. Therefore, it is necessary to find a technique for obtaining the hourly electricity demand from the peak demand provided by UNIDO. To accomplish this, a combination of the data obtained from a PhD research being undertaken at the University of Edinburgh on time-series demand model for Nigeria (Oluwale, 2018), some load demand profiles from literature and the personal observations made during the field visit to Nigeria, were used to generate the hourly load demand for each of the activities listed. Using the demand curves obtained from the above

Table 1
Unit capacity for activities and the total demands in case study communities (UNIDO, 2015).

Activities	Unit Capacity (kW)	Sabon Gari (kW)	Kakara (kW)	Bangoba (kW)	Kusuku (kW)	Lugere (kW)	Galadima (kW)	Nguroje (kW)	Furmi (kW)	Tunga (kW)	Chabal Gudali (kW)
Barbing/saloon shops	0.20	0.60	1.40	0.40	2.00	0.40	0.20	6.00	1.20	0.20	0.00
Sewing/fashion shops	0.20	0.60	2.40	0.80	2.40	0.20	0.20	8.00	0.60	0.00	0.20
Carpentry shops	0.20	0.00	1.60	0.00	0.60	0.20	0.60	0.20	0.00	0.00	0.00
Welding shops	2.00	2.00	4.00	2.00	6.00	2.00	2.00	30.00	4.00	0.00	0.00
Milling/grinding machines	0.50	1.00	4.50	1.50	4.50	0.50	2.00	7.50	4.00	0.50	0.50
Pepper/tomatoes grinders	0.20	0.20	0.20	0.00	0.20	0.00	0.40	4.00	0.00	0.00	0.00
Market area	0.30	0.30	0.30	0.00	0.30	0.00	0.30	0.90	0.30	0.00	0.00
Bakeries	2.00	0.00	0.00	0.00	2.00	0.00	0.00	10.00	0.00	0.00	0.00
Schools	0.25	0.25	1.00	0.25	1.25	0.25	0.25	2.50	2.50	0.25	0.00
Clinics/health centre	1.00	1.00	1.00	1.00	0.40	1.00	1.00	5.00	2.00	0.00	0.00
Viewing centre	0.20	0.20	0.20	0.40	1.00	0.00	0.20	3.40	0.80	0.00	0.00
Computer/business centre	0.20	0.60	0.40	0.00	6.00	0.00	0.20	2.00	0.80	0.00	0.00
Shops	0.30	0.00	9.60	0.90	0.00	1.20	1.80	45.00	9.00	0.00	0.00
Water pump	0.50	0.00	0.00	0.00	0.30	0.00	0.00	1.50	0.50	0.00	0.00
Musician (musical) appliances	0.30	0.60	0.30	0.30	0.30	0.30	0.30	0.90	0.60	0.30	0.30
Electrical iron, cooker & heater	2.00	30.00	50.00	20.00	12.00	6.00	10.00	56.00	10.00	2.00	0.00
TV set/compact disk/radio	0.15	2.25	15.75	1.50	2.25	1.20	1.05	18.00	5.25	0.60	0.30
Household lighting & fan	0.20	126.40	316.00	124.00	320.00	24.00	70.00	460.00	250.00	19.60	11.00
Churches	0.30	1.80	3.90	0.60	4.20	1.80	1.20	15.00	1.20	0.30	0.00
Filling station	0.80	0.00	0.80	0.00	0.00	0.00	0.00	4.80	0.00	0.00	0.00
Handset charging	0.10	0.10	0.20	0.10	0.30	0.10	0.10	1.20	0.30	0.10	0.00

sources as reference curves, a per unit scaling technique was used to obtain the demand curve for the activities in the case study communities. First, per unit values were estimated for each time t_1 to t_{24} using Eq. (9).

$$\text{Per unit value at time } t_n = \frac{\text{reference curve demand at time } t_n}{\text{reference curve Peak demand}} \quad (9)$$

The hourly demands for each activity in the study communities were obtained by multiplying the corresponding per unit value at time t_n with the study community peak demand. To obtain the actual demand, the result obtained for each hour is further multiplied by a demand factor of 0.7 which was obtained from the UNIDO data sheet. It was observed that the activities in all the case study communities are similar and variations are only a matter of scale.

2.5. The potential for demand management

It was observed that a significant percentage of the electricity demand came from lighting. It is largely responsible for the peak between 7 pm and 9 pm when electricity demand for Sabon Gari community is greater than 90 kW. However, for most hours of the day, the electricity demand for the community is below 20 kW. If the system sizing is done based on peak demand, the system will be greatly underutilized for most hours of the day. In order to deal with this, demand management options were considered. A study conducted by the Power Holding Company of Nigeria found that 85% of the light bulbs used in the country were incandescent light bulbs (The Power Holding Company of Nigeria, 2009). Hence, replacing incandescent light bulbs (assumed to be 85% of the light bulbs in this study) with LED bulbs will reduce the unit capacity for light bulbs and fan from the given value of 0.2 kW to 0.09375 kW (A logical assumption was made that the ratio of light bulbs to fans was 3:1). The impact of this change on the electricity demand profile of Sabon Gari community is shown in Fig. 5. It causes a 49.5% reduction in peak electricity demand. A similar trend is observed for the other 9 communities.

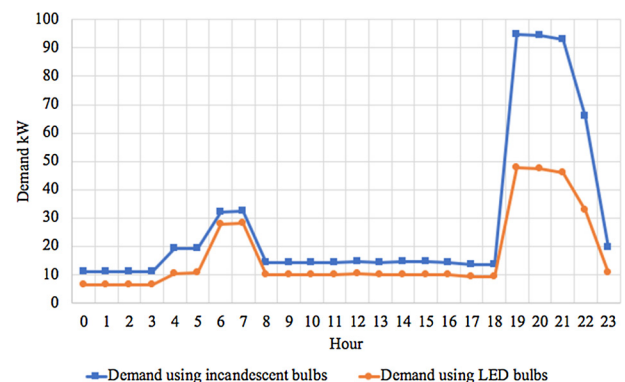


Fig. 5. Electricity demand profiles for Sabon Gari community.

2.6. Solar resource data

The solar resource data for Sabon Gari community (latitude 7.988 and longitude 10.993) were obtained from the European Commission Photovoltaic Geographic Information System. Due to the proximity of the 10 case study communities, the same solar resource data was used for all the communities.

2.7. Optimal sizing of system components using HOMER

The microgrid analysis version (HOMER pro) of the Hybrid Optimization Model for Electric Renewables was used for sizing the system components. Monthly solar irradiance and temperature data for the locations as well as hourly electricity demand were fed into the software. Generic flat plate PV panel, generic 12 V lead acid battery, diesel generator and converter were selected. The cost estimates provided by HOMER for the components are given in Table 2.

HOMER uses dispatch strategies to define the operation of the controller. The load following (LF) dispatch strategy was selected. In this strategy, the diesel generator is operated to serve only the load. In

Table 2
Costs, lifetime and efficiency of components.

Component	Capital cost (\$)	Operation & maintenance cost (\$)	Lifetime	Efficiency (%)
Generic flat plate PV	3 000/kW	10/kW/year	25 years	13
Generic lead-acid battery	300/kWh	10/kWh/year	10 years	80
Generic converter	300/kW	0	15 years	95
Diesel generator	500/kW	0.030/kW/hour	15 000 hours	–

contrast, the cycle charging strategy (CC) allows the diesel generator to operate at its rated capacity in order to serve the load and charge the battery.

The following optimization constraints were used: zero maximum annual capacity shortage, zero minimum renewable fraction and zero operating reserves (expressed as % of annual peak load). It was assumed that the depth of discharge of the battery was 100%, but in reality, the depth of discharge of batteries is usually less than 100%. A discount rate of 5.88% and project lifetime of 25 years was used. The fuel price for diesel was assumed to be \$1 per Litre. The information provided was used by HOMER to determine the component sizes that will satisfy the optimisation constraints at the lowest net present cost. The optimal system configurations for the case study communities were determined for two scenarios. The first scenario was the business-as-usual scenario where the communities use incandescent light bulbs. The second scenario assumes an energy efficient scenario where the communities use LED bulbs. Tables 3 and 4 below show the optimization results obtained from HOMER for the case study communities.

2.8. Simulations and verification of HOMER optimisation results

The optimal sizes of system components obtained from HOMER were inputted into the Simulink model initially developed. The hourly irradiance data and load for each community were fed into the Simulink model and the performance of the system over a 24-h period was observed. The Simulink model was also used to check the accuracy of sizes obtained from HOMER.

3. Results and discussion

3.1. Results

3.1.1. Analysis of system performance

Figs. 6 and 7 show the simulation results obtained for the business-as-usual scenario (incandescent light bulbs) and the improved energy efficiency scenario (LED bulbs) respectively for Sabon Gari. These results do not only show the performance of the system for a typical day but also indicates the accuracy or otherwise of the component sizes obtained from HOMER. The detailed graphical illustrations of the other 9 case study communities have not been presented here due to space constraint but are similar to Sabon Gari. It is evident from the results that no power was sent to the dummy load for both scenarios. This

Table 3
Microgrid component sizes for communities (incandescent light bulbs scenario).

Community	PV (kW)	Diesel Gen. (kW)	Battery (kWh)	Converter (kW)
Sabon gari	89.5	180	374	46.8
Kakara	247	470	906	102
Bangoba	82.3	180	329	39.6
Kusuku	238	460	771	87.6
Lugere	28.8	39.0	98	11.2
Galadima	63.1	110	221	28.4
Nguroje	629	780	2,091	225
Furmi	194	360	654	69.1
Tunga	12.7	28.0	45	4.86
Chabal Gudali	14.6	16.0	42	5.98

Table 4
Microgrid component sizes for energy efficient scenario (LED bulbs used).

Community	PV (kW)	Diesel Gen. (kW)	Battery (kWh)	Converter (kW)
Sabon gari	50	90	167	24.3
Kakara	169	250	550	72.3
Bangoba	50.9	85	183	26.3
Kusuku	156	240	536	58.6
Lugere	20.6	22.0	68	8.24
Galadima	32.8	25.0	96	11.3
Nguroje	475	450	1449	168
Furmi	130	190	417	45.6
Tunga	8.22	14.0	29	3.72
Chabal Gudali	10.4	7.50	17	5.07

confirms the accuracy of the optimal size of system components obtained from HOMER. A sensitivity test showed that increasing the PV size or decreasing the battery capacity caused a significant amount of power to be sent to the dummy load. Furthermore, if the size of the PV or battery is decreased, the power required from the diesel generator and the fuel consumption will increase significantly.

It was observed that the rated capacity of the diesel generator obtained from HOMER was high compared to the maximum output required from it for these simulations. However, this is because HOMER incorporated a safety factor in sizing the generator to ensure constant power supply for the community even when the PV is not generating any power due to bad weather or the load increases slightly beyond what was estimated. The same pattern is seen in the sizing of the diesel generator for the other communities in this study. From the results, it is seen that the diesel generator is operated in the early hours of the morning when the power generated from the PV is low. When the PV starts generating more power than is required by the load, the surplus is sent to the battery. As the PV generation starts declining in the evening hours and the PV is unable to meet the power requirements of the load, the energy stored in the battery is discharged to compensate for this deficit. The system performed as expected and the sizes of the components were found to be satisfactory.

In Kakara community, the system performance was similar to those of the Sabon Gari community. The system satisfied the hourly loads of the community without sending any power to the dummy load. Hence, the system performed as expected and the optimal sizes obtained from HOMER were found to be accurate. The simulation results obtained for Bangoba community show that both incandescent light bulbs and LED scenarios satisfied the hourly loads of the community without sending any power to the dummy load. The system performed as expected and the optimal sizes obtained from HOMER were confirmed to be accurate. In Kusuku community, for the business-as-usual scenario, the optimal component sizes obtained from HOMER were used for the simulations. The results show that excess power is sent to the dummy load at 3 pm when the state of charge of the battery is 100%. This makes it seem like the PV system was oversized or the battery was undersized. However, the simulation presented here was done using the hourly irradiance for a day in January. But HOMER optimises the system for a year using monthly average irradiance. Hence, the system was sized to ensure its reliability every day of the year. The simulation results for the LED bulbs scenario for Kusuku community show that the system satisfied the hourly loads of the community without sending any power to the

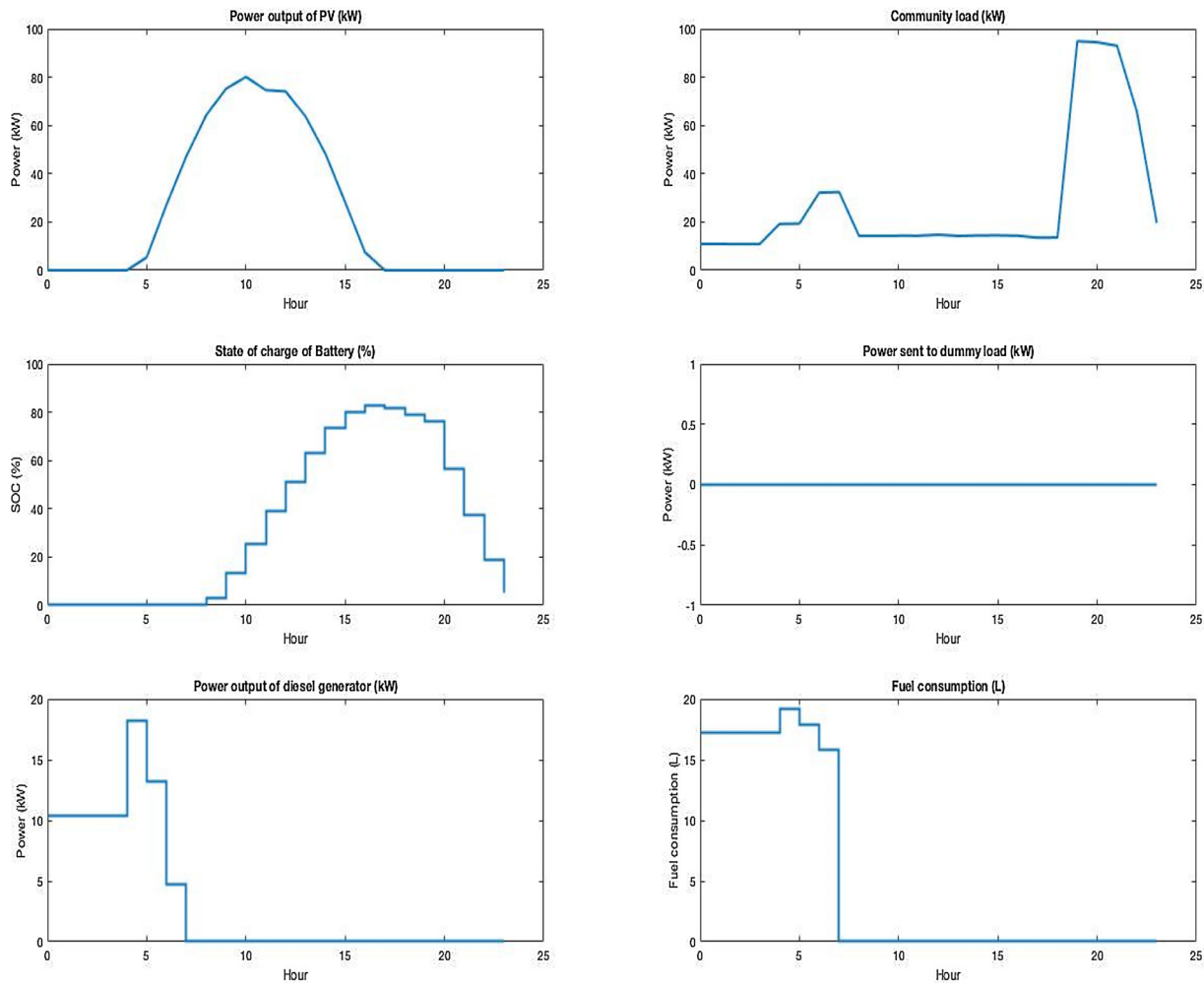


Fig. 6. Simulation results for Sabon Gari business-as-usual scenario (incandescent light bulbs).

dummy load. The system performed as expected and the optimal sizes obtained from HOMER were confirmed to be accurate.

The simulation results obtained for Lugere community for incandescent light bulbs scenario and LED bulbs scenario suggest that the system performed as expected for both scenarios and the optimal sizes obtained from HOMER were confirmed to be accurate since no power was sent to the dummy load. In Galadima, the simulation results obtained for the community in the business-as-usual scenario show that the system performed as expected and the optimal sizes obtained from HOMER for this scenario were found to be accurate. The simulation results for the improved energy efficiency (LED bulbs) scenario indicates that a small amount of power is sent to the dummy load at 3 pm. The results obtained for Nguroje community suggest that although a small amount of power is sent to the dummy load, however, this does not negate the accuracy of the component sizes obtained from HOMER. The system performance was satisfactory for both the incandescent light bulbs scenario and LED bulbs scenario.

The results obtained for Furmi community indicate that the system performance for both the incandescent light bulbs scenario and LED bulbs scenario was satisfactory. Similarly, in Tunga community, the system performed satisfactorily in both the incandescent light bulbs scenario and LED bulbs scenario. However, in Chabal Gudali community, a small amount of power is sent to the dummy load in both scenarios. For the LED bulbs scenario, the diesel generator is turned on to meet the load requirement when the state of charge of the battery is at 0%. Although increasing the battery capacity will make the operation of the diesel generator at this time unnecessary, but this might be a more

expensive option than using the diesel generator if this happens only for one day. HOMER's optimisation is based on selecting the option with the least NPC over the lifetime of the project.

3.1.2. Economic assessment

3.1.2.1. Comparison of LCOE. The LCOE is the ratio between the total discounted costs and discounted energy production over the lifetime of the project. HOMER was used to calculate the LCOE for four different scenarios for each of the case study communities. These scenarios are PV-battery-diesel generator combined with the use of LED bulbs, PV-battery-diesel generator combined with the use of incandescent light bulbs, diesel generator combined with the use of LED bulbs and lastly, diesel generator combined with the use of incandescent light bulbs. The LCOE per kWh for the 10 case study communities are shown in Fig. 8. The diesel generator-incandescent light bulbs scenario has the highest LCOE for all the case study communities while PV-battery-diesel generator-LED bulbs scenario has the lowest LCOE for most of the communities (on average it has 50% less LCOE than a system with only a diesel generator combined with the use of incandescent light bulbs).

3.1.2.2. Comparison of NPC. The NPC of the 4 scenarios for each of the communities is presented in Fig. 9. As expected, the scenario with PV, battery and diesel generator combined with the use of LED bulbs has the least NPC. The percentage difference between the NPC for PV-battery-diesel generator-LED bulbs scenario compared with using diesel generator-incandescent light bulbs scenario range between 56% (Nguroje) and 81% (Galadima).

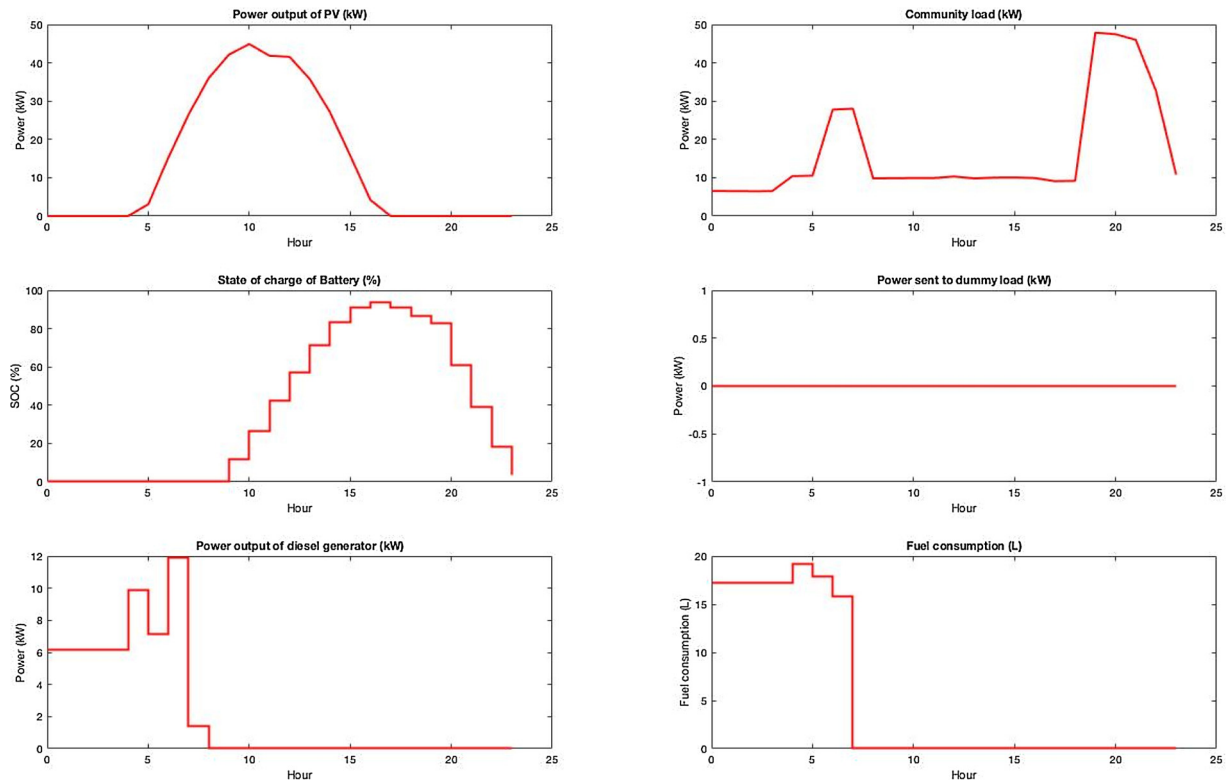


Fig. 7. Simulation results for Sabon Gari improved energy efficiency scenario (LED bulbs).

3.2. Discussion

It is evident from the results obtained that the PV, battery and diesel generator, combined with the use of LED bulbs is the most suitable scenario for all the communities since it has the least LCOE and NPC. Based on these results, a generalized conclusion can be drawn that the proposed configuration is suitable for rural areas in Nigeria while bearing in mind the underlying assumptions that may have influenced these results. It is important that while presenting the results, the underlying assumptions are also stated since they may have influenced the results obtained. Due to the proximity of the study communities, the same irradiance data was used for all the communities. Also, demand data from UNIDO and demand patterns from other research works were used and some logical assumptions were made while generating the demand profile. Furthermore, while developing the mathematical model for the smart microgrid, assumptions were made for some technical parameters such as the efficiency of components, the coefficients of fuel consumption of the diesel generator and temperature coefficient of the PV module. These assumptions were based on values obtained from other studies. Lastly, the economic analysis that produced the results for the LCOE and NPC was based on assumed values of capital cost, operation and maintenance cost, fuel cost and discount rate obtained from HOMER. These values may not accurately represent the current cost of the components. Also, the external cost such as the cost due to emissions from diesel generator was not included in the analysis. If there are inaccuracies in these data, they may have slightly influenced the validity of the results obtained.

The solar PV, battery and diesel generator smart microgrid configuration was found to be suitable for off-grid electrification in the case study communities. The proposed system has a greater economic advantage compared to a system using a diesel generator alone as indicated by a comparison of the LCOE and NPC. The proposed system will benefit rural communities in Nigeria who are unconnected to the main grid. The use of LED bulbs greatly reduced the electricity demand in the case study communities and also significantly reduced the net

present cost of the smart microgrid. Due to its low comparative cost, the proposed system has the potential to encourage private sector investments in off-grid rural electrification. Smart business models such as the pay-as-you-go model can be used to ensure that investors make a reasonable profit. In the pay-as-you-go business model, consumers can use prepaid meters. The use of smart meters and sensors in the smart microgrid will enable automatic detection of electricity theft thereby ensuring profitable investment. Since there is abundant solar energy resource in all parts of Nigeria, the proposed system will be suitable for electrification of rural areas in all parts of the country.

4. Conclusions and recommendations

The purpose of this research was to examine the potential of a smart microgrid for rural electrification in Nigeria. A solar PV, battery and diesel generator microgrid configuration was chosen. Mathematical modelling of the various components of the microgrid was carried out and a Simulink model of the smart microgrid system was developed. The electricity demands in 10 case study rural communities in Nigeria were analysed and demand profiles were generated. The possibility of demand management was considered. The optimal sizes of system components were found using HOMER for both business-as-usual scenario (incandescent light bulbs) and improved energy efficiency scenario (LED bulbs). The two scenarios were simulated using the model developed in Simulink using solar irradiance data for a typical day in January.

The system performance was found to be satisfactory and the component sizes obtained from HOMER met the requirement of the community electricity demand for both scenarios. The proposed system was found to have between 56% and 81% less NPC compared to a system with only diesel generator combined with the use of incandescent light bulbs. Also, the LCOE for the proposed system was found to be on average 50% less than the LCOE of a system with only a diesel generator combined with incandescent light bulbs. Hence, the proposed system has greater economic merit compared to a system with

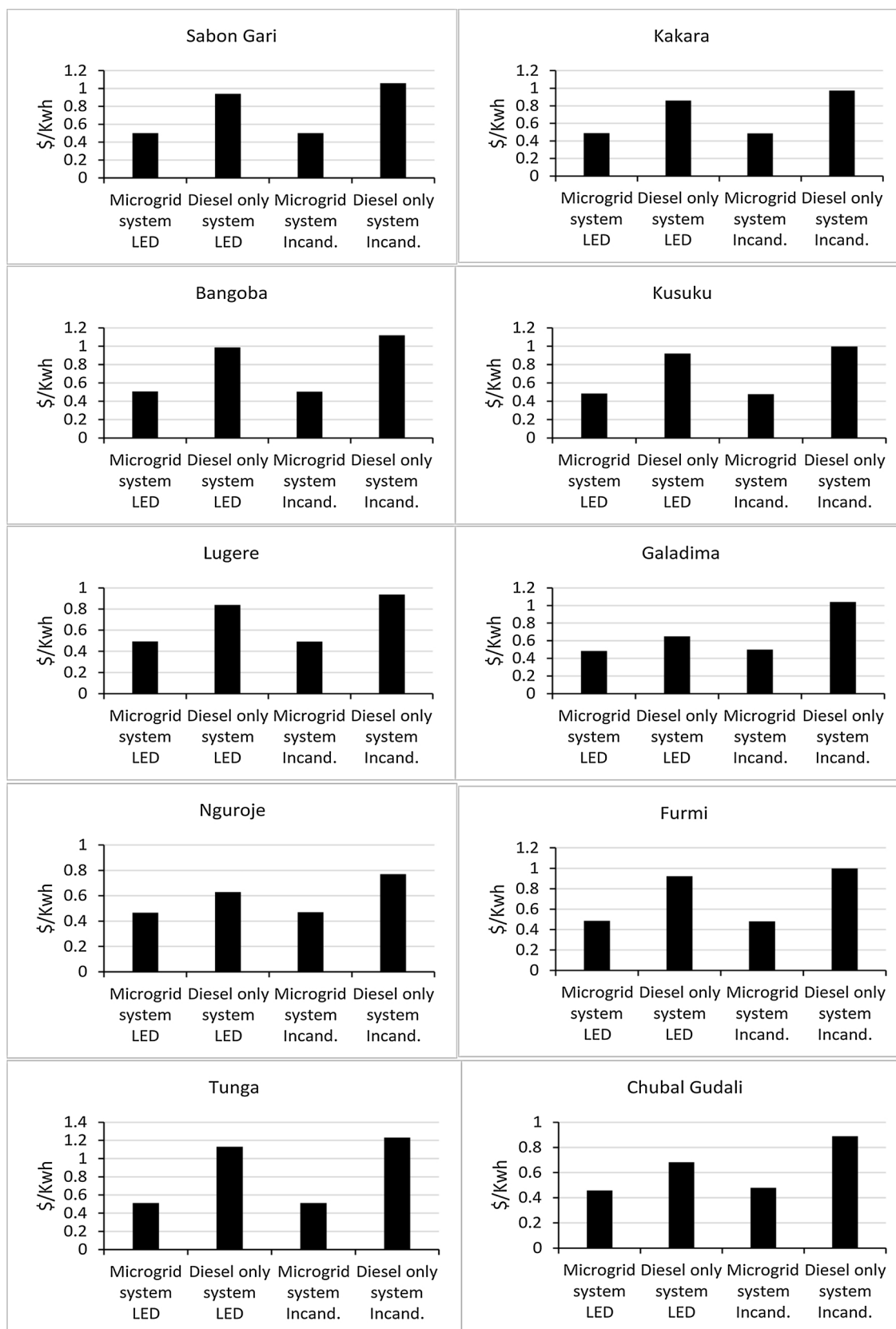


Fig. 8. LCOEs of case study communities.

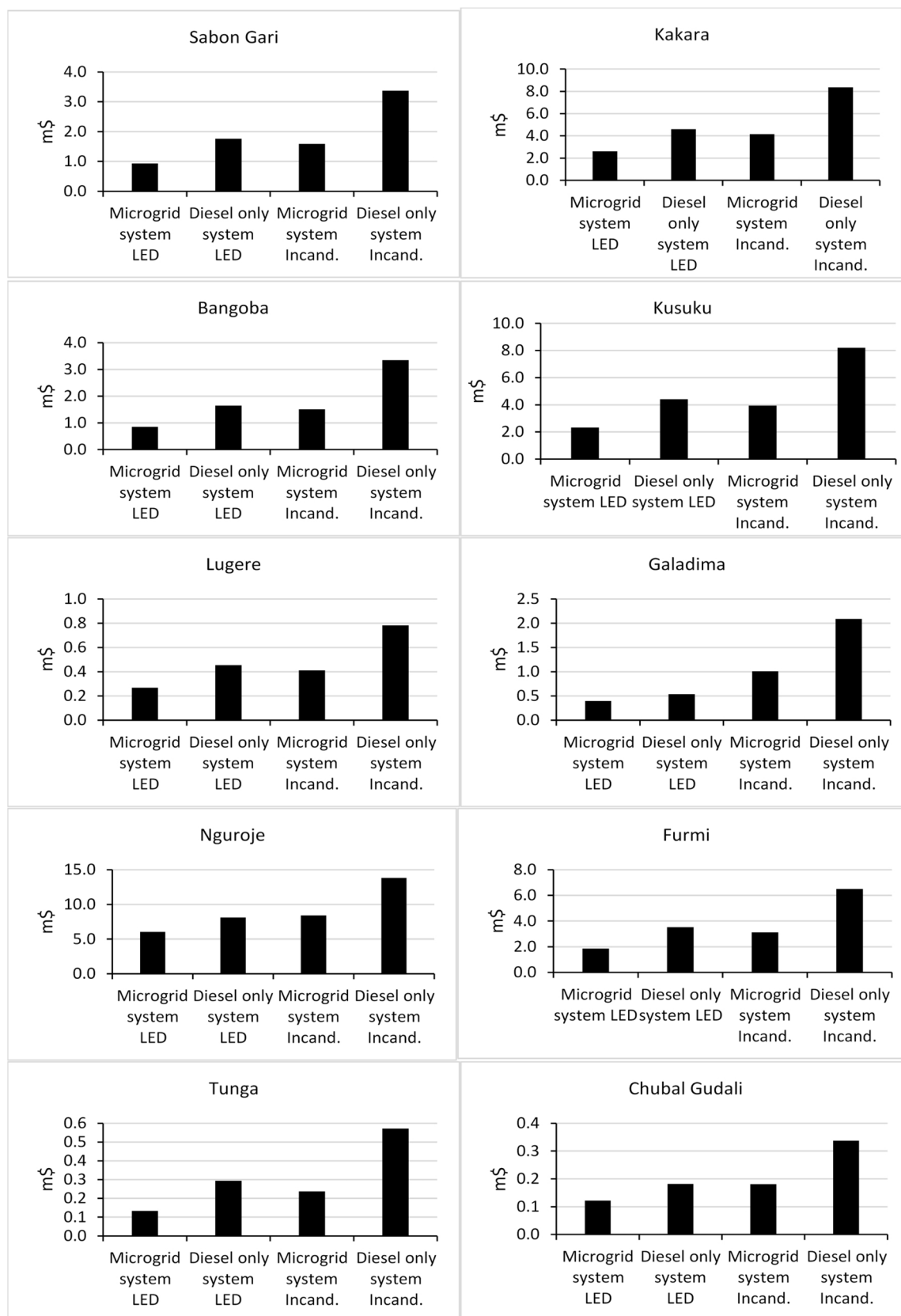


Fig. 9. NPCs of case study communities.

only diesel generators which are currently being used in many rural areas in Nigeria. The PV, battery and diesel generator smart microgrid combined with the use of LED bulbs is recommended for off-grid electrification of rural communities in Nigeria as it will ensure a reliable supply of electricity to these areas. Future work can be done to further validate the model using data from other communities as well as to propose policies that will facilitate the deployment of the system in the Nigerian rural electrification context.

Acknowledgements

We wish to express our gratitude to Dr Aristides Kiprakis, Dr Sasa Djokic, and Dr Damilola Oluwale of the College of Science and Engineering, the University of Edinburgh for providing guidance and support throughout this study. A big thank you to the Staff of the United Nations Industrial Development Organisation, Abuja for providing relevant data for this work.

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The Electricity Journal

Volume 33, Issue 4, May 2020, Page

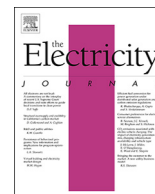
DOI: <https://doi.org/10.1016/j.tej.2020.106751>



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Corrigendum

Corrigendum to “Modelling and management of smart microgrid for rural electrification in sub-saharan Africa: The case of Nigeria” [Electric. J. 32 (2019) (10) 106672]



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The authors would like to apologise for any inconvenience caused.

DOI of original article: <https://doi.org/10.1016/j.tej.2019.106672>

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<https://doi.org/10.1016/j.tej.2020.106751>

Available online 26 March 2020

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